Active Disturbance Rejection Control (ADRC)

Dissolved oxygen regulation in the activated-sludge stage of a WWTP

Organized collection of equations

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1 Dissolved oxygen mass balance (plant)

Mass balance

$$\text{Accumulation} = \text{Input} - \text{Output} + \text{Generation} - \text{Consumption} \tag{1}$$

$$\frac{dS}{dt} = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{Q_{out}}{V}S + K_{L_a}\left(S^* - S\right) - r_z(S) \tag{2}$$

With the units of each term:

$$\underbrace{\frac{dS}{dt}}_{\frac{\text{mg}}{\text{L·h}}} = \underbrace{\frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i}}_{\frac{1}{\text{L}}\left(\frac{\text{L}}{\text{h}}\right)\left(\frac{\text{mg}}{\text{L}}\right)=\frac{\text{mg}}{\text{L·h}}} - \underbrace{\frac{Q_{out}}{V}S}_{\left(\frac{\text{L/h}}{\text{L}}\right)\left(\frac{\text{mg}}{\text{L}}\right)=\frac{\text{mg}}{\text{L·h}}} + \underbrace{K_{La}(S^* - S)}_{\left(\frac{1}{\text{h}}\right)\left(\frac{\text{mg}}{\text{L}}\right)=\frac{\text{mg}}{\text{L·h}}} - \underbrace{r_z(S)}_{\frac{\text{mg}}{\text{L·h}}} \tag{3}$$

With r_Z:

$$r_Z = -\frac{1 - Y_H}{Y_H}\,\rho_1 - \frac{4.57 - Y_A}{Y_A}\,\rho_3 \tag{4}$$

$$\rho_1 = \mu_H \left(\frac{S_S}{K_S + S_S}\right)\left(\frac{S_O}{K_{O,H} + S_O}\right) X_{B,H} \tag{5}$$

$$\rho_3 = \mu_A \left(\frac{S_{NH}}{K_{NH} + S_{NH}}\right)\left(\frac{S_O}{K_{O,A} + S_O}\right) X_{B,A} \tag{6}$$

1.1 Model chemical parameters (ASM1 / BSM1)

Parameter	Unit	Value
Y_A	g cell COD formed · (g N oxidized)$^{-1}$	0.24
Y_H	g cell COD formed · (g COD oxidized)$^{-1}$	0.67
f_P	dimensionless	0.08
i_{XB}	g N · (g COD)$^{-1}$ in biomass	0.08
i_{XP}	g N · (g COD)$^{-1}$ in particulate products	0.06

Table 1: Stoichiometric parameters

Parameter	**Unit**	**Value**
μ_H	d^{-1}	4.0
K_S	g COD $\cdot$ m^{-3}	10.0
$K_{O,H}$	g (-COD) $\cdot$ m^{-3}	0.2
K_{NO}	g NO_3-N $\cdot$ m^{-3}	0.5
b_H	d^{-1}	0.3
η_g	dimensionless	0.8
η_h	dimensionless	0.8
k_h	g slowly biodegradable COD $\cdot$ (g cell COD $\cdot$ d)$^{-1}$	3.0
K_X	g slowly biodegradable COD $\cdot$ (g cell COD)$^{-1}$	0.1
μ_A	d^{-1}	0.5
K_{NH}	g NH_3-N $\cdot$ m^{-3}	1.0
b_A	d^{-1}	0.05
$K_{O,A}$	g (-COD) $\cdot$ m^{-3}	0.4
k_a	$\mathrm{m}^3\cdot$ (g COD $\cdot$ d)$^{-1}$	0.05

Table 2: Kinetic parameters

2 The ADRC input gain: how b_1 (the gain that multiplies u) is obtained

In its canonical form, the ADRC models the plant as a chain of integrators plus a *total disturbance* f that groups all the unknown dynamics. Starting from the dissolved oxygen mass balance (handwritten note), the first-order plant is written as

$$\frac{dS_O}{dt} = b_0\, K_{L_a} - f, \qquad [f] = \frac{\text{mg}}{\text{L}\cdot t} = \frac{\text{mg}}{\text{L}}\cdot\frac{1}{t}, \tag{7}$$

and, more generally, for a higher-order system (the GPI form used in the design),

$$\boxed{\ddot{y} = b_1\, u - f} \tag{8}$$

Main idea. The extended state observer (ESO) estimates f online, so f does not need to be known. Instead, an estimate of the **input gain** b_1 (equivalently b_0, k or α) is required, because the control law inverts it:

$$u = \frac{1}{b_1}\big[-v(e_y, \dots) + \hat{f}\big]. \tag{9}$$

This gain **is not a single physical constant**: it is the *product of the static gains of the whole actuation chain* that turns the numeric controller output u into the real physical effect on oxygen (the transfer coefficient K_{L_a}). The handwritten note breaks this chain into four stages.

2.1 Actuation chain that builds b_1

1. PLC: 0–100% $\rightarrow$ 4–20 mA. The controller output (the PLC set-point in percent) is converted into a normalized analog current signal:

$$I = k_{\text{PLC}}\, u + 4 \text{ mA}, \qquad k_{\text{PLC}} = \frac{20-4}{100-0} = 0.16\ \frac{\text{mA}}{\%}. \tag{10}$$

2. Motor speed: n = 0–3000 rpm. The variable-frequency drive (VFD) maps the 4–20 mA current into the blower rotation speed:

$$n = k_{\text{VFD}}\,(I-4), \qquad k_{\text{VFD}} = \frac{3000-0}{20-4} = 187.5\ \frac{\text{rpm}}{\text{mA}}, \tag{11}$$

which, composed with the previous stage, is equivalent to

$$n = \frac{3000}{100}\,u = 30\,u \quad \left[\frac{\text{rpm}}{\%}\right]. \tag{12}$$

3. Air flow proportional to motor speed: Q_{air} [V/f] $\propto$ n. Under V/f control (blower affinity laws in the linear region), the air flow is proportional to speed:

$$Q_{\text{air}} = k_Q\, n, \qquad k_Q = \frac{Q_{\text{air}}^{\max}}{n^{\max}}. \tag{13}$$

4. Oxygen transfer coefficient proportional to air flow: K_{L_a} $\propto$ Q_{air}. Finally, the aeration delivers a transfer coefficient proportional to the injected air flow:

$$K_{L_a} = k_{K_{L_a}}\, Q_{\text{air}}. \tag{14}$$

2.2 Composition: the total gain

Chaining the four stages, the coefficient K_{L_a} becomes a linear function of u:

$$K_{L_a} = k_{K_{L_a}}\, k_Q\, k_{\mathrm{VFD}}\, k_{\mathrm{PLC}}\, u \;=\; \alpha\, u, \tag{15}$$

so the proportionality constant α (the gain that multiplies u) is

$$\boxed{\alpha = k_{K_{L_a}}\, k_Q\, k_{\mathrm{VFD}}\, k_{\mathrm{PLC}}} \tag{16}$$

and the ADRC input gain is

$$\boxed{b_1 = k = b_0\, \alpha = b_0\; k_{K_{L_a}}\, k_Q\, k_{\mathrm{VFD}}\, k_{\mathrm{PLC}}} \tag{17}$$

Stage	**Conversion**	**Gain**
1. PLC	0–100% $\rightarrow$ 4–20 mA	$k_{\mathrm{PLC}} = 0.16$ mA/%
2. VFD / motor	4–20 mA $\rightarrow$ 0–3000 rpm	$k_{\mathrm{VFD}} = 187.5$ rpm/mA
3. Blower V/f	$n \rightarrow Q_{\mathrm{air}}$	$k_Q = Q_{\mathrm{air}}^{\max}/n^{\max}$
4. Aeration	$Q_{\mathrm{air}} \rightarrow K_{L_a}$	$k_{K_{L_a}}$
Total gain $\alpha = K_{L_a}/u$		$\alpha = k_{K_{L_a}} k_Q k_{\mathrm{VFD}} k_{\mathrm{PLC}}$

Table 3: Decomposition of the input gain $b_1 = b_0\alpha$ along the actuation chain.

Note. In the simulation the whole chain is lumped into a single numeric value $\alpha = \frac{1}{30}$ h^{-1} per unit of u, that is

$$K_{L_a} = \frac{u}{30} \quad [\mathrm{h}^{-1}]. \tag{18}$$

3 ADRC canonical form and total disturbance

If the oxygen transfer coefficient is considered proportional to the control input,

$$K_{La} \propto u \tag{19}$$

then,

$$K_{La} = \alpha u \tag{20}$$

where α is a proportionality constant.
Substituting into the dissolved oxygen mass balance yields

$$\dot{S} = \underbrace{\frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{Q_{out}}{V}S - r_z}_{f(S,t)} + u \underbrace{\alpha\,(S^* - S)}_{g(S,t)}. \tag{21}$$

$$\dot{S} = f(S,t) + g(S,t)\,u \tag{22}$$

where

$$g(S,t) = \alpha\,(S^* - S) \tag{23}$$

and

$$f(S,t) = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{Q_{out}}{V}S - r_z(S) \tag{24}$$

All dynamics that differ from a pure integrator are grouped as a total disturbance f,

$$\dot{y} = f + b_0 \cdot K_{L_a} \tag{25}$$

where the total disturbance f is defined as:

$$f(S,t) = \frac{1}{V}\sum_{i=1}^{3} Q_{in,i}S_{in,i} - \frac{Q_{out}}{V}S - r_z(S) + K_{L_a}\,(S^* - S) - b_0 \cdot K_{L_a} \tag{26}$$

relationship between Oxygen transfer coefficient K_{La} and u. Oxygen transfer coefficient is proportional to the control signal u applied to the blowers:

$$K_{L_a} \propto u \tag{27}$$

$$K_{L_a} = \alpha \cdot u \tag{28}$$

with $k = b_0 \cdot \alpha$

$$\dot{y} = f + k \cdot u \tag{29}$$

4 ADRC design

Now, we have to design the ADRC. We rewrite the second-order system as a second-order error system. Let $e_y = y - r$:

$$\dot{e}_y = k \cdot u + (\bar{\xi} - u^*) \tag{30}$$

Defining $\xi = \bar{\xi} - u^*$:

$$\dot{e}_y = k \cdot u + \xi \tag{31}$$

Applying the same procedure as before, the estimation error dynamics become. We have to select the states of the system:

$$\dot{x}_1 = x_2 + k \cdot u \tag{32}$$
$$\dot{x}_2 = \text{PMI}(\xi, t) \tag{33}$$

Extended Dynamics System

$$\dot{X}_{au} = A_{au} \cdot X_{au} + B_{au} \cdot u + B_{\xi_{au}} \cdot \xi^{(m)} \tag{34}$$

with the state vectors defined as:

$$X_p = \left[e_y\right], \qquad X_\xi = \left[\xi\right] \tag{35}$$

and the augmented state:

$$X_{au} = \begin{bmatrix} X_p \\ X_\xi \end{bmatrix} \tag{36}$$

Observer System

$$\dot{\widehat{X}}_{au} = (A_{au} - LC_{au})\widehat{X}_{au} + B_{au}u + Le_y \tag{37}$$

with $e_y = C_{au} \cdot X_{au}$.

Estimation Error Dynamics

$$(\dot{X}_{au} - \dot{\widehat{X}}_{au}) = (A_{au} - LC_{au})(X_{au} - \widehat{X}_{au}) + B_{\xi_{au}}\xi^{(m)} \tag{38}$$

$$\dot{\bar{e}} = (A_{au} - LC_{au})\bar{e} + B_{\xi_{au}}\xi^{(m)} \tag{39}$$

The control law is given by the following expression, where the observer provides the estimated plant and disturbance states; with v:

$$u = k_1 \cdot \widehat{X}_{au_1} + k_2 \cdot \widehat{\xi} \tag{40}$$

$$u = K \cdot \widehat{X}_{au} \tag{41}$$

5 ADRC in discrete time

Disturbance Estimator

$$x[k+1] = A\,x[k] + b\big(u[k] + \xi[k]\big) \tag{42}$$

Disturbance system:

$$x_{im}[k+1] = A_{im}\,x_{im}[k] \tag{43}$$
$$\xi[k] = C_{im}\,x_{im} \tag{44}$$

Augmented system:

$$z[k+1] = A\,x[k] + b\,C_{im}\,x_{im}[k] + b\,u[k] \tag{45}$$

In matrix form:

$$\begin{bmatrix} x[k+1] \\ x_{im}[k+1] \end{bmatrix} = \underbrace{\begin{bmatrix} A & b\,C_{im} \\ 0 & A_{im} \end{bmatrix}}_{A_{ou}} \underbrace{\begin{bmatrix} x[k] \\ x_{im}[k] \end{bmatrix}}_{x_{au}} + \underbrace{\begin{bmatrix} b \\ 0 \end{bmatrix}}_{b_{au}} u \tag{46}$$

$$y = \underbrace{\begin{bmatrix} C & 0 \end{bmatrix}}_{C_{au}} \begin{bmatrix} x[k] \\ x_{im}[k] \end{bmatrix} \tag{47}$$

with Observer. Augmented system:

$$x_{au}[k+1] = A_{au}\, x_u + b_{au}\, u[k] \tag{48}$$

Observer equation:

$$\hat{x}_{au}[k+1] = A_{au}\, \hat{x}_{au}[k] + b_{au}\, u[k] + L_{au}\big(e[k] - \hat{e}[k]\big) \tag{49}$$

Expanded form:

$$\hat{x}_{au}[k+1] = \big(A_{au} - L_{au}\, C_{au}\big)\, \hat{x}_{au} + b_{au}\, u[k] - L_{au}\, e[k] \tag{50}$$

5.1 Control law (discrete)

$$u = \frac{1}{k_{sys}} \big[-v(e_y^{(0)}[k], \xi[k])\big] \tag{51}$$

$$u = \frac{1}{\kappa_{sys}} \left[-k_1\, \hat{x}_{ou_1}[k] + k_2\, \hat{x}_{ou}[k]\right] \tag{52}$$

$$u = \frac{1}{\kappa_{sys}} \Big[-k_1\, \hat{e}_y[k] + k_2\, \hat{\xi}[k]\Big] \tag{53}$$

6 Control law with GPI

Control law

$$u(s) = -\frac{1}{\kappa} \left[\frac{k_{n+m-1}s^{n+m-1} + \ldots + k_1 s + k_0}{s^m\,(s^{n-1} + k_{2n+m-2}s^{n-2} + \ldots + k_{n+m})}\right] e_y(s) \tag{54}$$

$$u(s) = -\frac{1}{\kappa} \left[\frac{k_1 s + k_0}{s\,(k_2)}\right] e_y(s) \tag{55}$$

6.1 Discrete time (Tustin)

Applying the Tustin discretization, the continuous operator s is replaced by:

$$s \;\longleftarrow\; \frac{2}{T_s} \cdot \frac{z-1}{z+1} \tag{56}$$

where T_s is the sampling period. Substituting into the controller:

$$u(z) = -\frac{1}{\kappa} \left[\frac{k_1 \left(\dfrac{2}{T_s} \cdot \dfrac{z-1}{z+1}\right) + k_0}{k_2 \left(\dfrac{2}{T_s} \cdot \dfrac{z-1}{z+1}\right)}\right] e_y(z) \tag{57}$$

Simplifying:

$$u(z) = -\frac{1}{\kappa\, k_2} \left[\frac{\left(\dfrac{2k_1}{T_s} + k_0\right) z + \left(k_0 - \dfrac{2k_1}{T_s}\right)}{\dfrac{2}{T_s}(z-1)}\right] e_y(z) \tag{58}$$

In difference equation form:

$$u[k] = u[k-1] - \frac{T_s}{2\,\kappa\,k_2}\left[\left(\frac{2k_1}{T_s} + k_0\right) e_y[k] + \left(k_0 - \frac{2k_1}{T_s}\right) e_y[k-1]\right] \tag{59}$$

7 Example: Discrete-Time DC Motor

This block models stage 2 of the actuation chain (Section 2): the servomotor that sets the blower speed. Its own ADRC/observer is designed in the same way.

7.1 Motor transfer function

The motor transfer function is given by

$$G(s) = \frac{\theta(s)}{U(s)} = \frac{k_\omega}{\tau s + 1} \cdot \frac{k_\theta}{s} = \frac{k}{s(\tau s + 1)}. \tag{60}$$

where k_ω and k_θ are gains for unit conversion, for example, ω in rev/s and θ in degrees.
A state-space representation with

$$x_p(t) = \begin{bmatrix} \theta(t) \\ \omega(t) \end{bmatrix} \tag{61}$$

results in

$$\dot{x}_p(t) = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{1}{\tau} \end{bmatrix} x_p(t) + \begin{bmatrix} 0 \\ \frac{k}{\tau} \end{bmatrix} u(t), \tag{62}$$

and

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x_p(t). \tag{63}$$

After discretization we obtain

$$x_p(k+1) = A_p x_p(k) + B_p u(k), \tag{64}$$
$$y(k) = C_p x_p(k). \tag{65}$$

7.2 Two integral (internal model)

The internal model is described by

$$x_\xi(k+1) = A_\xi x_\xi(k) + B_\xi \, \xi^{(m)}(k), \tag{66}$$
$$\xi(k) = C_\xi x_\xi(k). \tag{67}$$

7.3 Augmented system (extended system)

The augmented discrete-time system is

$$\begin{bmatrix} x_p(k+1) \\ \xi(k+1) \end{bmatrix} = \begin{bmatrix} A_p & B_p C_\xi \\ 0_{(m,n)} & A_\xi \end{bmatrix} \begin{bmatrix} x_p(k) \\ \xi(k) \end{bmatrix} + \begin{bmatrix} B_p \\ 0_{(m,1)} \end{bmatrix} u(k) + \begin{bmatrix} 0_{(n,1)} \\ B_\xi \end{bmatrix} \xi^{(m)}(k). \tag{68}$$

The output equation is

$$y(k) = \begin{bmatrix} C_p & 0_{(1,m)} \end{bmatrix} \begin{bmatrix} x_p(k) \\ \xi(k) \end{bmatrix}. \tag{69}$$

In compact form,

$$x_{au}(k+1) = A_{au} x_{au}(k) + B_{au} u(k) + B_{\xi,au} \, \xi^{(m)}(k), \tag{70}$$
$$y(k) = C_{au} x_{au}(k). \tag{71}$$

where the augmented state vector is

$$x_{au}(k) = \begin{bmatrix} x_p(k) \\ \xi(k) \end{bmatrix}. \tag{72}$$

7.4 Observer system

The discrete-time observer is

$$\widehat{x}_{au}(k+1) = (A_{au} - LC_{au})\,\widehat{x}_{au}(k) + B_{au}u(k) + Le_y(k). \tag{73}$$

where

$$e_y(k) = C_{au}x_{au}(k). \tag{74}$$

Estimation error dynamics. Defining the estimation error as

$$\bar{e}(k) = x_{au}(k) - \widehat{x}_{au}(k), \tag{75}$$

its dynamics are

$$\bar{e}(k+1) = (A_{au} - LC_{au})\,\bar{e}(k) + B_{\xi,au}\,\xi^{(m)}(k). \tag{76}$$

7.5 State feedback control law

The control law is

$$u(k) = -k_1\widehat{x}_{au,1}(k) - k_2\widehat{x}_{au,2}(k) - \cdots - k_{n+m}\widehat{x}_{au,n+m}(k), \tag{77}$$

or, in compact form,

$$u(k) = -K\widehat{x}_{au}(k). \tag{78}$$

Since

$$\widehat{x}_{au}(k) = \begin{bmatrix} \widehat{x}_p(k) \\ \widehat{\xi}(k) \end{bmatrix}, \tag{79}$$

the control law can also be written as

$$u(k) = -K_p\widehat{x}_p(k) - 1 \cdot \widehat{\xi}(k). \tag{80}$$

7.6 Closed-loop augmented system

The closed-loop dynamics are

$$\begin{bmatrix} x_{au}(k+1) \\ \bar{e}(k+1) \end{bmatrix} = \begin{bmatrix} A_{au} - B_{au}K & B_{au}K \\ 0 & A_{au} - LC_{au} \end{bmatrix} \begin{bmatrix} x_{au}(k) \\ \bar{e}(k) \end{bmatrix}. \tag{81}$$

7.7 Closed-loop system for simulation

The discrete-time closed-loop system is

$$\begin{bmatrix} x_p(k+1) \\ \widehat{x}_{au}(k+1) \end{bmatrix} = \begin{bmatrix} A_p & -B_pK \\ LC_p & A_{au} - LC_{au} - B_{au}K \end{bmatrix} \begin{bmatrix} x_p(k) \\ \widehat{x}_{au}(k) \end{bmatrix} + \begin{bmatrix} 0_{(n,1)} & B_p \\ -L & 0_{(n+m,1)} \end{bmatrix} \begin{bmatrix} r(k) \\ \xi(k) \end{bmatrix}. \tag{82}$$

7.8 Observer difference equations

$$\hat{x}(k+1) = (A - LC)\,\hat{x}(k) + Bu(k) + Le_y(k) \tag{83}$$

$$\hat{x}(k) = (A - LC)\,\hat{x}(k-1) + Bu(k-1) + Le_y(k-1) \tag{84}$$

8 Design and simulation numerical parameters

Symbol	Description	Value
Dissolved oxygen plant (ADRC_ASM)		
T_s	sampling period	0.05 h (3 min)
k_p	design gain (integrator k_p/s)	$2/3\ \mathrm{h}^{-1}$
α	$K_{L_a} = \alpha\, u$	1/30
S^*	O_2 saturation at 15 °C	8.0 mg/L
Servomotor / actuator (ADRC_v2)		
M_p	target overshoot	25%
e_v	velocity error	0
e_a	acceleration error	0.001
m_f	target phase margin	40°
m_g	target gain margin	10 dB
T_s	sampling period	0.005 s
K_θ	angular conversion	360
$G(s)$	identified plant	$2201/(s + 45.08)$

Table 4: Numerical constants used in the design and simulation.